

Executive Summary

The Middle East and North Africa remain the world's hydrocarbon heartland, hosting the vast majority of global oil and gas reserves [90†L1268-L1271] [73†L4098-L4106] . About 80% of world oil reserves lie in OPEC Middle East countries (Saudi 267×10^9 bbl, Iraq 145×10^9 bbl, Iran 209×10^9 bbl)

[90†L1268-L1271] [74†L372-L374] , and a similar share of gas reserves (Qatar $\sim 24 \times 10^{12} \text{ m}^3$, Iran $\sim 34 \times 10^{12} \text{ m}^3$, UAE $\sim 8.2 \times 10^{12} \text{ m}^3$) [73†L4098-L4106] [74†L372-L374] . Most production today still comes from giant fields in the Arabian/Persian Gulf (Ghawar, Safaniya, Rumaila, etc.) and North African basins (Sirte, Hassi Messaoud), with smaller newer production in the Levant and East Mediterranean (e.g. Leviathan, Zohr). Many sedimentary basins have been intensely explored, but significant frontier potential remains in under-explored provinces (deepwater Red Sea, Levantine offshore, NW Africa Atlantic margin, Zagros deep plays). Recent large gas discoveries (Egypt's Zohr, Israel's Leviathan/Tamar, Mauritania-Senegal "Tortue") have reshaped regional gas prospects [98†L170-L174] [105†L800-L808] . Looking ahead, the most prospective areas include Eastern Mediterranean and NW Africa for gas, and deeper/deformed flanks of Gulf and Red Sea basins for oil. Key risks are geopolitical (sanctions, conflicts, border disputes) and infrastructure bottlenecks. Technology trends – deepwater drilling, unconventional (shale/tight) prospects, and CCS projects (e.g. Saudi Uthmaniyah, UAE Habshan) – are reshaping exploration. (See Table below for reserves/production and Mermaid timeline for major discoveries.)

Regional Geology & Resources

Arabia-North Africa's petroleum systems are dominated by Mesozoic–Paleozoic carbonate reservoirs in the Arabian Plate and Cretaceous–Tertiary clastics in peripheral basins. The Persian Gulf "Arabian Basin" (Kuwait, Saudi, UAE, Qatar, Iraq, Iran) contains supergiant Paleozoic oil fields (e.g. Ghawar, Safaniya) and associated giant gas (e.g. North Gas Field) [74†L372-L374] [93†L4107-L4115] . Westward, the Zagros fold-thrust belt (Iraq and Iran) holds enormous oil and some gas (fields Rumaila, Kirkuk, South Pars gas). North African platforms (Libya, Egypt, Algeria) have Triassic–Jurassic and Cretaceous reservoirs in basins like Sirte, Gulf of Suez and Nile Delta, supporting Libya and Algeria's multi-billion-barrel fields [90†L1250-L1258] . The Levantine/Nile–Eastern Mediterranean (Levant) basins are gas-rich: Zohr (Egypt) and Tamar/Leviathan (Israel) proved huge gas (Zohr $\sim 30 \times 10^{12} \text{ m}^3$) [105†L800-L808] [98†L170-L174] , but host little oil. Overall the region's proven oil reserves are $\sim 1.57 \times 10^{12}$ bbl (end-2024) [72†L331-L335] [90†L1268-L1271] and gas $\sim 209 \times 10^{12} \text{ m}^3$ [73†L4098-L4106] [90†L1268-L1271] . The Persian Gulf holds by far the largest share (e.g. Saudi Arabia $\sim 267 \times 10^9$ bbl oil [74†L372-L374] , plus $102.9 \times 10^9 \text{ m}^3$ gas production in 2024 [74†L378-L381]).

Country Highlights

- **Saudi Arabia** – World's largest oilholder ($\sim 267 \times 10^9$ bbl [74†L372-L374]) and producer ($\sim 12.7 \times 10^3$ kb/d in 2024 [44†L264-L268]). Key basin: Rub' al Khali and eastern Arabian Basin; giant fields Ghawar, Safaniya. Gas reserves $\sim 9.7 \times 10^{12} \text{ m}^3$ [74†L372-L374] (88% associated).
- **Iraq** – Proven oil $\sim 145 \times 10^9$ bbl [74†L372-L374] , producing $\sim 3.9 \times 10^3$ kb/d [74†L375-L377] (2019–2023 growth halted by OPEC quotas). Basins: Mesopotamian Foredeep (Rumaila, West Qurna). Gas reserves $\sim 3.7 \times 10^{12} \text{ m}^3$ [74†L374-L376] . Challenges: infrastructure and Figs pipeline.

- **Iran** – $\sim 209 \times 10^9$ bbl oil reserves [74†L372-L374] , $\sim 3.8 \times 10^3$ kb/d production. Huge gas ($\sim 34 \times 10^{12}$ m³) in South Pars (shared field) and Zagros. Sanctions limit investment, but geology still prolific (Karun/Wagheh).
- **Kuwait** – $\sim 101 \times 10^9$ bbl [74†L372-L374] , $\sim 2.4 \times 10^3$ kb/d. Burgan field (2nd largest world) and deeper Jurassic plays (Risasi/Shuaiba). Gas $\sim 1.8 \times 10^{12}$ m³ [74†L374-L376] (gas production $\sim 16.3 \times 10^9$ m³/yr in 2024 [74†L378-L381]).
- **UAE** – Abu Dhabi $\sim 113 \times 10^9$ bbl [74†L372-L374] , 2.9×10^3 kb/d. Onshore basins (e.g. Shah, Upper Zakum) and offshore (Umm Shaif). Gas $\sim 8.2 \times 10^{12}$ m³ [74†L374-L376] (2024 prod $\sim 57.8 \times 10^9$ m³ [74†L378-L381]). ADNOC expanding recovery and Dubai's Fateh field offshore.
- **Qatar/Bahrain** – Qatar $\sim 25 \times 10^9$ bbl oil (mainly condensate) plus world's 3rd-largest gas (23.8×10^{12} m³). Bahrain minor oil (0.126×10^9 bbl) and mature fields (Bahrain, Awali). Qatar's North Field underpins LNG expansion, Bahrain has heavy Qusaib oil.
- **Oman** – $\sim 4.8 \times 10^9$ bbl [76†L65-L68] , ~ 775 kb/d [76†L59-L62] (heavy oil from central fields: Marmul, Yibal). Gas $\sim 0.64 \times 10^{12}$ m³ [73†L4100-L4104] . Major tight gas (Khazzan field ~ 10 Tcf) and emerging discoveries (OQ North Oman).
- **Egypt** – $\sim 3.3 \times 10^9$ bbl [90†L1250-L1252] , $\sim 0.51 \times 10^3$ kb/d [92†L1969-L1972] . Key basins: Nile Delta, Western Desert (Bahrayne, Burullus fields), Gulf of Suez. Zohr (2015) revealed $\sim 30 \times 10^{12}$ m³ gas [105†L800-L808] ; Egypt's gas reserves $\sim 2.21 \times 10^{12}$ m³ [93†L4109-L4116] (2024 prod $\sim 63.6 \times 10^9$ m³ [96†L1-L4]). Strong recent exploration (several offshore finds beyond Zohr).
- **Libya** – $\sim 48.4 \times 10^9$ bbl [90†L1252-L1255] , $\sim 0.13 \times 10^3$ kb/d [92†L1970-L1973] . Sirte Basin fields (Sharara, Waha) are largest. Gas $\sim 0.73 \times 10^{12}$ m³ [93†L4112-L4116] (2024 prod $\sim 16.4 \times 10^9$ m³ [95†L1-L4]). Output is volatile due to conflict.
- **Algeria** – $\sim 12.2 \times 10^9$ bbl [90†L1249-L1252] , $\sim 0.70 \times 10^3$ kb/d [92†L1967-L1970] . Hassi R'Mel (gas) and Hassi Messaoud (oil) are major fields. Gas is main hydrocarbon (4.50×10^{12} m³ reserves [93†L4108-L4115] , $\sim 277 \times 10^9$ m³/yr prod [74†L378-L381]). (Maps of global basins and tables of reserves/production are provided below.)

Basin-Level Geology and Play Types

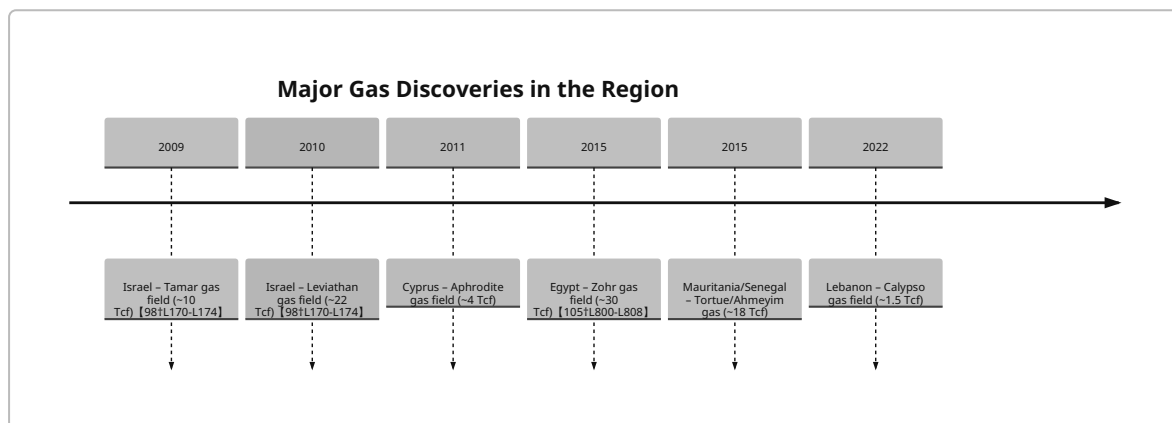
- **Persian Gulf/Arabian Shelf** – Thick Paleozoic-Khuff carbonates and Triassic/Jurassic reservoirs. Extremely mature (Ghawar, Safaniya, Burgan) but still some undeveloped deeper plays. "Intrashelf" slopes and Silurian shale cap are key.
- **Zagros Thrust Belt (Iran/Iraq)** – Folded basement-uplift province. Cretaceous–Jurassic carbonate thrust traps (Kirkuk, Ratawi) and Triassic–Permian sands. Large deep-well potential underplanted by Jurassic source. Highly prospective but complex geopolitics (Iran) and infrastructure constraints (northern Iraq).
- **Red Sea Rift** – Extensional rift basins offshore Egypt/Sudan/Saudi. Tertiary clastics with shallow (10–20 km) sub-basins. Recent seismic shows potential deep horizons; Chevron's Sidri Field (Egypt) proved Miocene oil. Underexplored due to drilling challenges.
- **Nile Delta & Suez** – Offshore/ onshore Cenozoic deltas and Upper Cretaceous carbonate (West Delta Deep Marine fields). Zohr in offshore Shorouk (Eocene carbonate) is a giant gas discovery [105†L800-L808] . Subsalt Miocene plays are frontier.
- **North Africa Saharan Platform (Sirte, Berkine, Murzuq)** – Triassic clastics (Nubian sandstone) and Miocene carbonates. Libya's Sirte Basin has many giants, Algeria's Berkine (Illizi) basin many smaller oil fields. Tunisia-Morocco Paleozoic/Pliocene plays are mostly gas-challenged.
- **Eastern Mediterranean/Levant** – Deepwater Miocene and Upper Cretaceous reservoirs under thick salt. Proven gas success (Zohr, Leviathan) but oil disappointing. Frontier drilling continues offshore Cyprus, Lebanon (Calypso gas ~ 1.5 Tcf found in 2022). Territorial disputes and immature infrastructure are key risks.

- **Atlantic Margin (West Africa/Mauritania)** – Passive margin deepwater turbidites. Recent BP “Tortue/Ahmeyim” finds (2015) revealed ~18 Tcf gas. Geologically similar to Guyana. Exploration still incipient.

Top Prospective Areas (Oil & Gas)

- **Eastern Mediterranean Gas** – Unconventional Miocene plays offshore Cyprus/Lebanon (giant gas shows Leviathan, Apophis). Remaining undrilled prospects (Cyprus Blocks 6,8; Lebanon Blocks 4,8) rank high, though disputed claims and lack of export infrastructure pose political risk [98†L170-L174] .
- **NW Asia/Red Sea Oil** – The northern Red Sea flanks (Egypt/Saudi) may host trapped Jurassic/Triassic oil. Saudi Ghawar’s western extension and NE Yemen failed rift basins (Hadramawt) also rank high. However, border conflicts (Yemen war) and high exploration costs are risk factors.
- **Subsalt Nile/Egypt Oil** – Extended Miocene carbonates under salt (Zohr analog) are largely undrilled; prospectivity is proven gas, with some oil potential. State incentives and drilling technology (deepwater techniques) reduce risk.
- **Deep Zagros (Iran/Iraq) Oil** – Undeveloped Jurassic-Cretaceous plays in thrust belt foothills. Very large upside (e.g. West Qurna expansions) but high geological complexity and political/sanctions risk.
- **Kuwait/Saudi Jurassic Oil** – “Hot shale” shales and fractured Arab formations west of existing fields, as hinted by Kuwait’s Jurassic gas (Darif) and Saudi Shaybah discoveries. Technically challenging but large payoffs if successful.
- **North-West Africa Gas** – Mauritania’s offshore gas (Basin C, now producing via TAN) and Moroccan Atlantic margin (Safi, Tendirra) remain high upside, balanced by frontier exploration risk and fledgling pipelines.

Recent Discoveries (Timeline)



Exploration Trends & Outlook

Investment and techniques are evolving. **Unconventionals:** Only limited shale/tight production exists (e.g. Oman’s Khufai and Saudi rig work on Permian analogues). Middle East shales (Ghawar shale, Early Cretaceous sands) remain under-tested. **Deepwater/High Pressure:** The success of deep Mediterranean and Atlantic deepwater (Zohr, Leviathan, Tortue) has spurred a drilling boom offshore, aided by better seismic imaging and drilling tech. **Decarbonisation/CCS:** Gulf states lead in CCUS; Saudi’s Uthmaniyah gas plant injects CO₂, Abu Dhabi’s Al Reyadah (Emirates Steel) is a first mover. These enable enhanced oil recovery (e.g. Aramco/EOR projects). **Investment Outlook:** Despite climate pressures, MENA governments (with large budgets) continue allocating capital to upstream (e.g. Iraq

exploration rounds, Iran’s recent oil field bids, Egypt’s offshore tenders). However, global energy transition and volatile prices introduce uncertainty. Countries are balancing near-term hydrocarbon development with longer-term diversification.

Table: Key Reserves & Production (end-2024 data, with major country/field context)

Country	Proven Oil Reserves (billion barrels) 【Source】	Oil Prod (kb/d, 2024) 【Source】	Proven Gas Reserves (billion m³) 【Source】	Gas Prod (billion m³/yr, 2024) 【Source】
Saudi Arabia	267.2 【74†L372-L374】	12,700 【44†L264-L268】	9,727 【74†L374-L376】	102.9 【74†L378-L381】
Iraq	145.0 【74†L372-L374】	3,862 【74†L375-L377】	3,714 【74†L374-L376】	11.1 【74†L378-L381】
Iran	208.6 【74†L372-L374】	(sanctions)	33,988 【73†L4098-L4104】	(N/A)
UAE	113.0 【74†L372-L374】	2,916 【74†L375-L377】	8,210 【74†L374-L376】	57.8 【74†L378-L381】
Kuwait	101.5 【74†L372-L374】	2,411 【74†L375-L377】	1,784 【74†L374-L376】	16.3 【74†L378-L381】
Oman	4.8 【76†L65-L68】	775 【76†L59-L62】	644 【73†L4099-L4102】	6.9 【74†L378-L381】
Egypt	3.3 【90†L1250-L1252】	512 【92†L1969-L1972】	2,209 【93†L4111-L4116】	63.6 【96†L1-L4】
Libya	48.4 【90†L1252-L1255】	132 【92†L1970-L1973】	730 【93†L4114-L4116】	16.4 【95†L1-L4】
Algeria	12.2 【90†L1249-L1252】	703 【92†L1967-L1970】	4,504 【93†L4107-L4112】	277.6 【74†L378-L381】

Data gaps/notes: Non-OPEC figures (e.g. Morocco, Sudan) are modest; some data (e.g. Syria, South Sudan) are unreliable. Many cited figures come from industry reports and country disclosures 【74†L372-L374】 【90†L1250-L1255】 , but true “proven” reserves may differ under different definitions. Exploration maturity varies: Gulf fields are highly mature, while deepwater/lean plays are frontier.

Sources: BP Statistical Review (via OPEC ASB) 【90†L1268-L1271】 , OPEC Annual Statistical Bulletin 2025 【74†L372-L374】 【93†L4107-L4115】 , national sources (e.g. Oman MEM) 【76†L65-L68】 , Eni/industry reports 【105†L800-L808】 【98†L170-L174】 . These are combined with IEA/EIA analysis and recent geology literature.